

Ph.D. RESEARCH PROPOSAL

Doctoral Program in Architecture

Breaking the Aerodynamic Short-Circuit: A Flow-Physics Account and Decoupled Resolution of the Sensible-Latent Conflict in Integrated Passive Envelopes for High-Enthalpy Mid-Rise Offices

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1. BACKGROUND AND PROBLEM STATEMENT

Decarbonizing the global commercial building stock is now among the most urgent task in the built environment. Building account for roughly 40% of global energy demand and a large share of energy-related CO₂ emissions, and intensifying heatwaves are pushing offices worldwide into ever-heavier reliance on mechanical air-conditioning – a feedback loop in which cooling demand and carbon emissions escalate together. Reducing the cooling load of commercial buildings without surrendering the indoor environment is therefore a shared, cross-regional challenge, not a local one.

Passive envelope strategies – double-skin façades (DSF), windcatcher, and external louvers – are widely promoted as the sustainable response to this challenge. Yet the prevailing literature evaluates these technologies largely in isolation, under single-performance metrics, or in temperate climates, and overwhelmingly treats the buoyancy-driven DSF cavity as a mechanism that enhances natural ventilation. Localized recirculation and flow reversal within the DSF cavity have, in fact, been documented in a number of experimental and numerical studies (e.g. Malgaard et al., 2023), yet largely as isolated flow-structure observations in single-façade configurations under temperate conditions.

The humid tropics provide the most severe test of these assumptions, and for that reason serve as the **extreme boundary condition** for this research than as a merely regional concern. A constantly high-enthalpy climate (in Jakarta, dry-bulb 27 – 36 °C at roughly 80% relative humidity, with no seasonal relief) subjects a passive envelope to a year-round combined sensible-and-latent load that temperate-climate studies never impose. Under these conditions a second, less-recognized failure emerges: the same openings a passive envelope requires for ventilation simultaneously draw humid outdoor air into the occupied zone, converting a ventilation strategy into a **latent-load liability**.

Summertime humidity management is a well-known constraint on passive design well beyond the tropics, which makes the tropical case a physically informative upper bound: a mechanism explained where the latent load is most punishing is a mechanism understood everywhere it matters. This coupling – where optimizing the envelope for sensible cooling actively worsens both fresh-air delivery and latent load – is the physics this research sets out to explain and resolve.

2. THE MASTER'S FINDING – DETECTING THE SHORT-CIRCUIT

My master's research directly the prevailing ventilation assumption. Using a two-phase parametric framework coupling EnergyPlus with an Airflow Network (AFN) diagnostic, applied to a six-storey Jakarta office archetype across 32 configurations, it found that an integrated passive envelope – a deep-cavity DSF, a windcatcher empirically recharacterized as a **Thermal Stack Tower (TST)**, and external louvers – reaches a hard thermodynamic limit.

The heated façade generated a parasitic exhaust that monopolized the building's aerodynamics: internal circulation appeared active (**0.392 ach**), but genuine fresh outdoor air captured by the TST collapsed to 0.073 ach – an 89.3% deficit against the ASHRAE 62.1 minimum (**0.68 ach**), with 81.4% of internal air movement being parasitic recirculation rather than outdoor-air exchange. The cavity exhausted 5.41x more air than the TST supplied, stalling fresh-air delivery despite the tower's design capacity. This study termed this the **Aerodynamic Short-Circuit Ratio (ASR)**. Crucially, optimizing sensible cooling (a 26.47% total-EUI reduction, to 110.14 kWh/m²/year against a 149.80 kWh/m²/year baseline) simultaneously triggered a latent energy penalty, as the ventilation openings admitting outdoor air drew tropical moisture into the occupied zone.

This research thus detected an aerodynamic failure and proposed – but did not test – a “decoupled hybrid” remedy. Two questions remain open, and they are precisely the questions an airflow network cannot answer: why the short-circuit occurs at the flow-field level (AFN averages inter-zone flow and cannot resolve the velocity and pressure field that produces recirculation), and how the envelope geometry can be re-engineered to break the short-circuit while preserving each component's function and resolving the sensible-latent conflict.

3. RESEARCH AIM, QUESTIONS, AND HYPHOTHESIS

Aim. To explain the aerodynamic mechanism that limits integrated passive envelopes under high-enthalpy conditions, and to develop and validate a re-engineered, decoupled envelope framework (THADES) that resolves the sensible-latent conflict for hot-humid mid-rise offices.

***Overarching question:** How can the aerodynamic short-circuiting of integrated passive envelopes in high-enthalpy climates be mechanistically and resolved to realize an energy-efficient decoupled (THADES) framework??*

RQ1 – MECHANISM

What flow-physics mechanism causes the DSF cavity to monopolize airflow and short-circuit TST-driven fresh-air delivery, and how does it manifest spatially in ways an AFN cannot capture?

Hypothesis: the buoyancy-driven cavity exhaust establishes a low-pressure region that re-entrains supply air, producing recirculation instead of throughflow. Method: high-resolution CFD (steady and transient), validated against the master’s AFN results and published benchmark datasets.

RQ2 – RESOLUTION & OPTIMIZATION

Which geometric and configurational interventions break the short-circuit while preserving solar shading (louver), buoyancy exhaust (TST), and thermal buffering (DSF)?

Method: CFD-based multi-objective optimization. Because direct CFD evaluation across a high-dimensional geometric design space is computationally prohibitive in both time and cost, machine-learning surrogate models are trained on a structured sample of CFD runs to approximate performance and make the optimization tractable – turning an otherwise intractable search into an algorithmically efficient framework. This computational efficiency is not incidental; it is what makes flow-resolved optimization of an integrated envelope feasible at all.

RQ3 – VALIDATION OF THE DECOUPLED FRAMEWORK

To what extent does the optimized THADES decoupled configuration resolve the sensible-latent conflict identified at master’s level, and how robust is the diagnostic method beyond the primary case?

Method: integrated coupled simulation and validation, with a bounded generalization study across representative humid-tropical urban climates (e.g. Jakarta, Kuala Lumpur, Singapore) as the closing phase. The tropical case is treated as the extreme upper bound: because the mechanism is explained and resolved where the latent load is most severe, the resulting ASR diagnostic and CFD-surrogate methodology are positioned as a transferable for subsequent investigation of other humid-load context, including humid continental summers, rather than assumed to generalize to them without further study.

4. SIGNIFICANCE & INTENDED CONTRIBUTION

This research project targets contributions on three levels:

- **Mechanistically**, while the existence of façade short-circuiting is itself acknowledged in the literature, this research offers the first flow-field-resolved account of *why* it arises in integrated

passive envelopes under high-enthalpy conditions – advancing the field from noticing the failure, and even detecting it quantitatively at the network scale (master's, via AFN), to resolving its underlying physics (PhD, via CFD).

- **Methodologically**, the contribution is not a better but a generalizable predictive tool: the ASR metric and the THADES framework are developed to mature from single-case constructs into a validated instrument that couples machine-learning surrogate modelling with computational fluid dynamics. This integration of AI-accelerated surrogates with flow physics constitutes a transferable methodological asset with potential applications for other researchers and practitioners. The ASR is offered as a candidate diagnostic metric for the wider building-performance-simulation community for identifying aerodynamic short-circuiting in any integrated passive system.
- **In applied and policy terms**, the work escalates from regional guidance to internationally recognized performance criteria. My master's finding quantified an 89.3% fresh air deficit against ASHRAE 62.1 - -an international ventilation standard – exposing a gap that current codes do not test: natural-ventilation credit is granted without explicit treatment of latent load. The ASR metric provides exactly the diagnostic such a verified-delivery criterion would need. The findings therefore inform not only regional standards (SNI, GBCI-GREENSHIP_ but the broader international conversation on how ventilation performance should be evidenced in codes and rating system.

The methods proposed (CFD, multi-objective optimization, and machine learning) are the means of answering RQ1-RQ3, justified by the questions they resolve rather than adopted for their own sake.

5. METHODOLOGY & RESEARCH DESIGN

The program is organized in three phases mapping onto RQ1 → RQ2 → RQ3, building directly on the validated master's foundation (EnergyPlus with AFN, a Jakarta EPW climate file, and six-storey mid-rise Office archetype).

Phase 1 – Mechanism (RQ1)

High-resolution CFD is used to resolve the cavity and TST flow field that the AFN can only average. The work begins by reconstructing the master's archetype as a CFD domain, then progressively resolving the buoyancy-driven cavity exhaust and its interaction with TST-driven supply. Validation follows a tiered strategy: numerical verification through grid-independence and convergence testing; validation against published experimental and CFD benchmark datasets; and cross-validation against the master's AFN results. Where laboratory of field-measurement access can be secured, experimental

validation will be incorporated to strengthen confidence; the design does not depend on it as a single point of failure.

Phase 2 – Resolution & Optimization (RQ2)

Rather than re-optimizing the prior study’s variable (master degree) set at network-averaged resolution, this study work re-examines the envelope’s geometric variables with high-resolution CFD, which – unlike the AFN – can resolve the cavity- and inlet-scale flow that governs aerodynamic capture and short-circuiting. **TST inlet configuration is added as a new variable as new variable not examined in the prior study, since only high-resolution CFD can resolve the inlet-scale flow relevant.** Because direct CFD evaluation across the design space is computationally demanding, machine-learning surrogate models are trained on a structured sample of CFD runs to approximate performance and enable multi-objective optimization against the competing objectives of fresh-air delivery, sensible cooling load, latent penalty, and daylight provision. The surrogate’s accuracy is verified before use (e.g. via RMSE against held-out CFD cases), and a variance-based sensitivity analysis (Sobol indices) is then applied to identify which variables and interactions dominate flow performance – it is this analysis, **rather than any assumption made in advance, that determines which variables drive the optimization and which may be fixed.** All simulation orchestration, surrogate training, and sensitivity analysis are scripted in Python, establishing a reproducible computational pipeline.

Phase 3 – Decoupled Framework Validation and Generalization (RQ3)

The optimized envelope configuration from phase 2 is integrated with the active mechanical supplementation prescribed by the THADES framework (such as a dedicated outdoor-air system or low-energy exhaust within the TST shaft) and evaluated through coupled simulation. This constitutes the first implementation and test of the THADES decoupled framework, which has so far been proposed only as a conceptual design paradigm and not yet evaluated, quantifying the extent to which it resolves the sensible-latent conflict it was formulated to address. Where time and computational scope permit, the phase closes with an optional bounded generalization study across two to three representative tropical urban climates (e.g. Jakarta, Kuala Lumpur and Singapore), to test the transferability of both the framework and the ASR metric beyond the primary case.

6. INDICATIVE TIMELINE

Year	Focus	Key Deliverables
Year 1	Coursework (where required); literature synthesis; CFD competency development; model build + numerical verification	Validated CFD model; RQ 1 mechanism paper draft

Year 2	Core CFD simulations; ASR mechanism analysis; surrogate-model development	RQ1 publication; optimization pipeline
Year 3	Multi-objective optimization; optimized envelope configuration; begin generalization	RQ2 publication
Year 4	Integrated validation; generalization study; synthesis & dissertation	RQ3 publication; doctoral thesis

Detailed first-year milestones to be expanded; later years intentionally broader, consistent with research-degree planning conventions

7. CANDIDATE PREPARATION

I am prepared to contribute as an independent researcher from the outset. My master's work required to build and validate the full EnergyPlus-AFN methodology, formulate the ASR diagnostic, and diagnose the aerodynamic failure this doctoral project now sets out to explain – therefore I begin with genuine command of the problem, the archetype, and the THADES construct rather than borrowed brief. My hard skill set is already aligned with the computational demands of the proposed work: variance-based sensitivity analysis using Sobol indices, Python scripting for simulation orchestration and data analysis, and the methodological transition from network-level modelling (EnergyPlus-AFN) to resolved fluid mechanics (CFD) with machine-learning surrogate acceleration. Eight years of professional practice as an architect on tropical commercial projects complements this computational foundation by keeping the work grounded in the realities of the building stock it aims to improve, but it is the computational capability that positions me to work productively in a building-science laboratory from the first year.

8. SELECTED REFERENCES

Selected from the master's verified reference base (foundational to the doctoral work); to be expanded with CFD, surrogate-modelling, and tropical-ventilation during full development.

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